Organic Farming
https://www.acadlore.com/journals/OF

Sustainable and Organic Farming Practices for Achieving the 2030 Sustainable Development Goals: A Case Study of Wheat Cultivation in Western Uttar Pradesh, India

Sonika Tyagi*, Yogesh Kumar Gupta

Department of Computer Science, Faculty of Mathematics and Computing, Banasthali Vidyapith, 304022 Banasthali, India

* Correspondence: Sonika Tyagi (21sonikatyagi@gmail.com)

Received: 04-24-2026 **Revised:** 06-22-2026 **Accepted:** 06-26-2026

Citation: Tyagi, S. & Gupta, Y. K. (2026). Sustainable and organic farming practices for achieving the 2030 Sustainable Development Goals: A case study of wheat cultivation in Western Uttar Pradesh, India. *Org. Farming*, *12*(2), 131–139. https://doi.org/10.56578/of120205.

© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: As the 2030 Sustainable Development Goals (SDGs) deadline approaches, only a small percentage of targets are on track. Agriculture cuts across at least six SDGs—hunger, health, water, climate, land, and responsible consumption—making it an important tool for course correction by supporting economic stability and environmental protection. In India, a fast-growing digital agricultural infrastructure, smart farming, and organic or natural farming are increasingly being integrated, whereas some conventional farming practices can degrade arable land and undermine long-term food security. Therefore, more sustainable farming approaches are required to meet future food demands. This paper proposes a conceptual framework of sustainable and organic farming techniques across the crop lifecycle, using a wheat case study in Western Uttar Pradesh, India, a leading wheat-producing region. The paper synthesizes sustainable-agriculture methods from the literature and situates them in this illustrative case, using secondary data and reported agronomic characteristics rather than primary field measurements. It also presents a notional smart-farming component for sensor-based irrigation scheduling. The framework distinguishes between SDGs 2, 6, 7, 12, 13, and 15, which the proposed practices directly target, and those that may be indirectly affected. Finally, the paper describes the quantitative, field-based validation needed to empirically evaluate the framework.

Keywords: Sustainable agriculture; Organic farming; Wheat cultivation; Sustainable Development Goals; Smart irrigation; Western Uttar Pradesh

1. Introduction

Agriculture is positioned as a central pathway for achieving multiple Sustainable Development Goals (SDGs) at the same time, including SDG 2 (Zero Hunger), SDG 3 (Good Health and Well-being), SDG 6 (Clean Water and Sanitation), SDG 8 (Decent Work and Economic Growth), SDG 12 (Responsible Consumption and Production), SDG 13 (Climate Action), and SDG 15 (Life on Land). At the midpoint of the 2030 timeline, however, global development has been judged as significantly off-track, with reporting at the 2023 G20 Summit indicating that just about 12% of SDG targets were on pace for timely achievement. This gap has increased interest in agricultural transformation pathways that can bring benefits across multiple goals at the same time, rather than focusing solely on output or revenue.

The SDGs for 2030 are founded upon five fundamental principles, known as the "5Ps" (United Nations Economic and Social Commission for Western Asia, 2018):

- **People:** Eliminating hunger and poverty in all manifestations, while guaranteeing that each person can achieve their utmost potential in a healthy setting, with equity and dignity.
- **Planet:** Safeguarding the Earth from deterioration through the sustainable stewardship of natural resources, curtailing unsustainable consumption and production practices, and promptly addressing climate change.
- **Prosperity:** Guaranteeing that all individuals can experience a thriving and satisfying existence, while

ensuring that economic, social, and technical advancements align with environmental sustainability.

- **Peace:** Fostering harmonious, equitable, and inclusive communities devoid of violence and apprehension.
- **Partnership:** Harnessing the necessary resources to implement the 2030 Agenda through cooperative efforts rooted in solidarity, with a specific focus on the requirements of the most disadvantaged.

The 17 SDGs were established based on these "5Ps", with a particular focus on food security. The SDG 2 clearly urges the eradication of hunger, attainment of food security, improvement of nutrition, and promotion of sustainable agriculture (United Nations, 2019). Sustainable agriculture includes various techniques, practices, and technological innovations that promote this goal (Hiywotu, 2025). and seeks to protect and maintain natural resources (Kiełbasa et al., 2018). Realizing this goal by 2030 necessitates a profound overhaul of the global food and agricultural framework (United Nations, 2019), with agricultural sustainability now widely acknowledged as a complex aim that directly supports certain SDGs and indirectly influences numerous others (Chimi et al., 2025).

Traditional agricultural practices, which depend significantly on chemical fertilizers, have contributed to production stability (Kumar & Manshi, 2023) but have also generated severe externalities that hinder sustainable development, such as nutrient depletion, greenhouse gas emissions, farming-related pollution, biodiversity decline, forest depletion, and degradation of arable land, as illustrated in Figure 1. Given that present agricultural production seems unsustainable in the long run (Suji & Kumar, 2020), it is essential to secure a sufficient food supply in the future, prompting a transition to sustainable, environmentally friendly farming practices (Vats, 2025). Climate-resilient agricultural methods that diminish dependence on pesticides and synthetic fertilizers are being progressively promoted to enhance soil vitality and preserve biodiversity (Chiriacò et al., 2017; Lee & Yun, 2015; Page et al., 2020; Rizzo et al., 2024). Embracing sustainable, environmentally friendly technologies and practices can foster enduring beneficial transformation (Schnack et al., 2024) by: (i) improving agricultural output to bolster food security and overarching developmental objectives; (ii) augmenting resilience and adaptability to climate change while preserving biodiversity and nutrient integrity; and (iii) reducing greenhouse gas emissions, pollution, and soil degradation. An environmentally sustainable approach can thus improve agricultural output without harming the ecosystem, establishing it as a strategic imperative for enduring sustainability and food security.

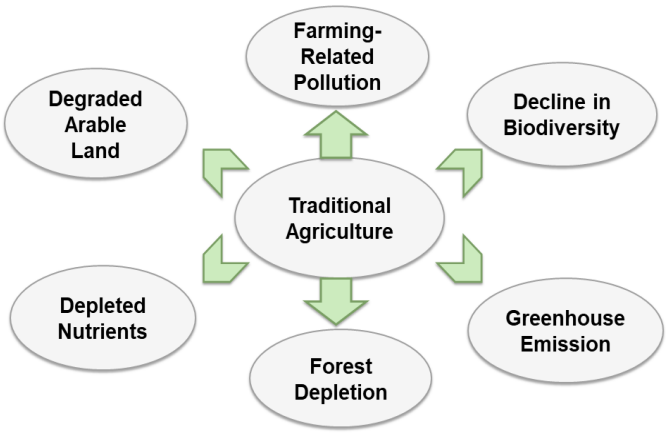


Figure 1. Ecological consequences of traditional agriculture

This study develops a conceptual framework that integrates established sustainable and organic farming practices across the crop lifecycle and illustrates their practical application through a wheat case study in Western Uttar Pradesh, India.

2. Methods

Inconsistent agricultural practices, coupled with continued global population growth, have placed considerable pressure on food security. The Global Hunger Index (GHI) score was 18.3 in 2025, indicating a moderate level of hunger worldwide (Scherer et al., 2025). However, progress toward eliminating hunger has stalled, placing the goal of achieving Zero Hunger by 2030 increasingly out of reach. Addressing these challenges requires a greater focus on sustainable agricultural practices. As economies transition from linear to circular models, green technologies play a pivotal role in promoting environmental sustainability and economic viability (Asiedu-Ayeh et al., 2022; Ikram et al., 2021).

The study framework is illustrated in Figure 2. It was developed through a narrative synthesis of recognized sustainable and organic agricultural practices across eight phases of the crop lifecycle, including crop selection, land preparation, seed selection, manure and fertilization, seed sowing, irrigation, harvesting, and storage. The framework was then demonstrated through an illustrative case study of wheat cultivation in Western Uttar Pradesh,

India.

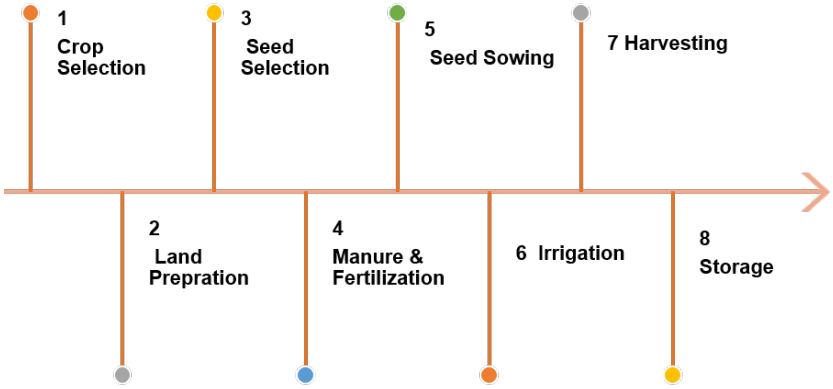


Figure 2. Significant milestones in the farming crop cycle

Uttar Pradesh was selected for the case study because of its significant agricultural output. Within the state, the Western Plain Zone exhibits high agricultural productivity and has a large proportion of irrigated land, approximately 89.23%. Therefore, the Western Plain Zone was selected as the geographical focus of the illustrative case study. The region has two main agricultural seasons, Rabi and Kharif, and the present study focuses on the Rabi season. Wheat is a major crop cultivated during this season; therefore, wheat cultivation in the Western Plain Zone of Uttar Pradesh was selected for the illustrative case study. To provide regional agricultural context, Figure 3 presents the cumulative production of 14 selected Rabi-season crops across seven districts in Western Uttar Pradesh (Meerut, Baghpat, Muzaffarnagar, Ghaziabad, Gautam Buddha Nagar, Bulandshahr, and Saharanpur) during the 2020–2021, 2021–2022, and 2022–2023 agricultural years, based on district-level data obtained from the Directorate of Economics and Statistics, Government of India.

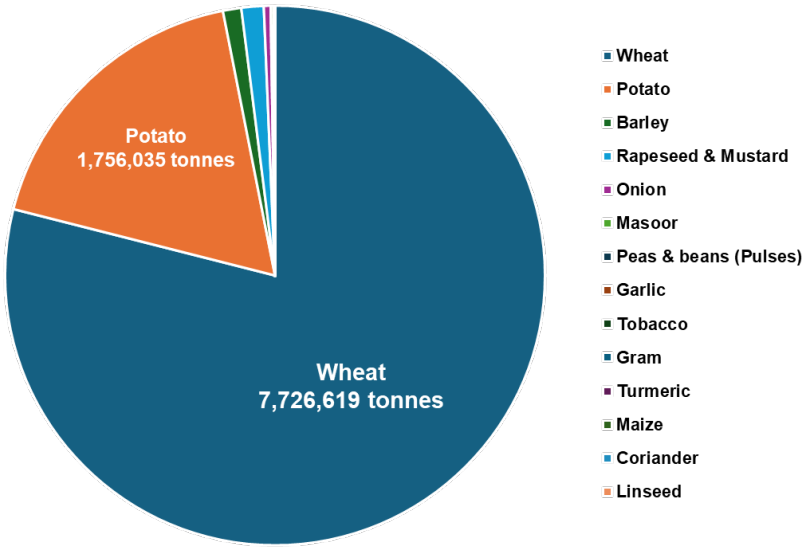


Figure 3. Distribution of cumulative production among 14 selected Rabi-season crops across seven districts in Western Uttar Pradesh, India, during the 2020–2021, 2021–2022, and 2022–2023 agricultural years
Source: Authors' calculations based on district-level crop production data from the Directorate of Economics and Statistics, Department of Agriculture and Farmers Welfare, Ministry of Agriculture and Farmers Welfare, Government of India. Crops–Area, Production and Yield (APY) Database. Available at: https://data.desagri.gov.in/website/crops-apy-report-web.

For each phase of the agricultural lifecycle, general sustainable practices are first outlined, followed by their application to the wheat case study in Western Uttar Pradesh.

(a) Crop selection

Crop selection is a critical step that influences subsequent phases of the agricultural lifecycle. The principal considerations include:

- Opting for regionally sourced crops minimizes energy-intensive logistics and storage, curtails carbon emissions, and enhances the local economy.

- Opting for crops that are compatible with local water supply.
- Employing rotational cropping to maximize nutrient cycling, diminish pest and disease proliferation, and elevate soil health.
- Merging several crops to create a more durable and diverse farming framework.

Application to the wheat case study: Based on the above criteria, the predominant Rabi-season cropping patterns identified in Western Uttar Pradesh were rice–wheat, bajra–wheat, sorghum–wheat, and sugarcane–wheat. Wheat was therefore selected as the Rabi/winter crop for this case study.

(b) Land preparation

Appropriate land preparation can improve long-term soil health by reducing erosion, improving soil structure, supporting weed management, and facilitating water and nutrient availability. It also provides suitable conditions for crop growth and can reduce reliance on unsustainable practices such as excessive chemical use.

Wheat case study: Land preparation involved plowing with a rotavator, followed by two passes with a disk harrow to obtain a finer soil texture. Initially, 15–25 kg of nitrogen (in the form of urea) per hectare was applied to facilitate the decomposition of crop residues in the soil. Furrows were created using a bed planter. Prior to planting, the field was graded with a uniform slope to ensure even water distribution.

(c) Seed selection

Key factors to consider include:

- Selecting indigenous, seasonal, and climate-adapted cultivars with high nutritional value.
- Using pure, certified, and appropriately treated seeds or sourcing seeds from local seed banks that conserve genetic diversity.
- Choosing climate-adaptive seeds suited to local water availability.
- Selecting seeds appropriate for organic agriculture to enhance soil health and reduce chemical inputs.
- Prioritizing heirloom and landrace seeds over hybrids, where appropriate, to maintain biodiversity.
- Using improved varieties that can enhance productivity, agricultural income, and food security.

Wheat seed selection: For wheat seed selection, the wheat varieties reported by the ICAR–Indian Institute of Wheat and Barley Research (n.d.) were evaluated using the proposed algorithm, primarily based on productivity and time to harvest.

Algorithm 1. Graph_Harvest_Data()

```
BEGIN
    // Step 1: Connect to Database
    Connect to MongoDB collection "seed_ranking"
    // Step 2: Group Records by Condition
    FOR each unique condition DO
        Calculate:
            Average harvest time (avg_harvest)
            Average productivity (avg_productivity)
            Average plant height (avg_height)
            Total number of records (count)
    END FOR
    // Step 3: Prepare Output
    Create output containing:
        labels          ← List of conditions
        avg_harvest     ← Average harvest values
        avg_productivity ← Average productivity values
        avg_height      ← Average height values
        count           ← Number of records
    // Step 4: Return Result
    RETURN the output in JSON format
END
```

The ranking results for wheat varieties based on average productivity and shortest harvest time are presented in Figure 4. After applying the above algorithm, DBW-17 was identified as the selected variety, with its main agronomic characteristics presented in Table 1.

Table 1. Agronomic characteristics of the selected wheat variety DBW-17

Seed Type	Crop Sown Time	Crop Harvest Time (Days)	Productivity (q/ha)	Plant Height (cm)	Irrigation Condition
DBW-17	Second fortnight of November	125–130	60–65	95–100	Irrigated

Note: q = quintal (100 kg); ha = hectare.

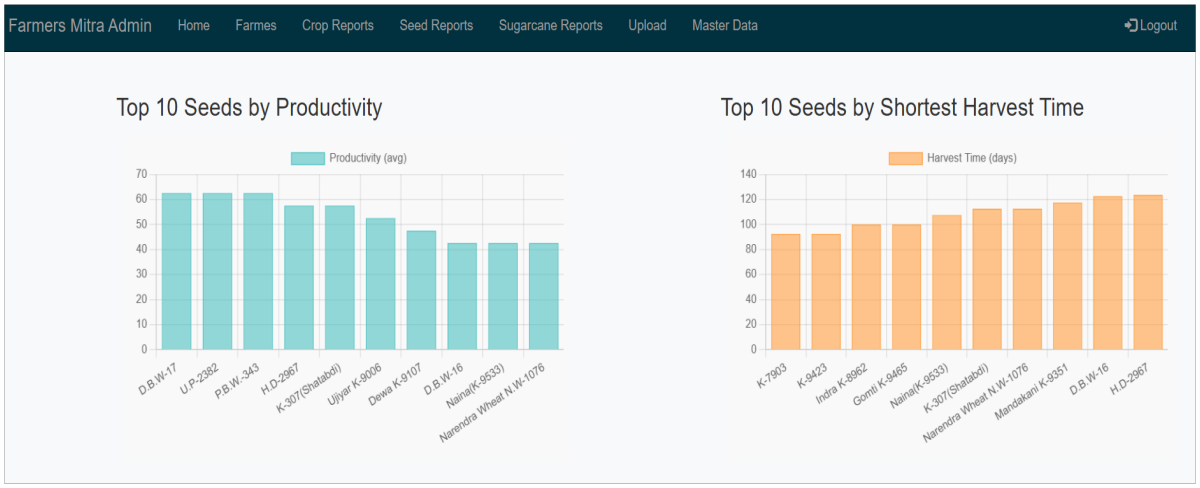


Figure 4. Ranking of wheat varieties by productivity and harvest time, reproduced from Tyagi & Gupta (2026) under the Creative Commons CC BY 4.0 license

(d) Manure and fertilizer

Recommended strategies include selecting fertilizers based on soil analysis; prioritizing organic fertilizers, biofertilizers, compost, manure, green manure, and biochar to improve soil structure and nutrient availability while reducing reliance on synthetic fertilizers; applying nutrients at appropriate rates, times, and locations according to soil conditions and crop requirements; adopting crop rotation to improve soil structure, nutrient cycling, and pest management; and recovering nutrients from urban waste streams, such as struvite, as a potential substitute for mined phosphorus.

Application to DBW-17 wheat: Fertilizer requirements depend on soil conditions and the availability of micronutrients and macronutrients. The micronutrient and macronutrient status of soils in the Western Plain Zone of Uttar Pradesh is presented in Table 2.

Table 2. Micronutrient and macronutrient status of soils in the Western Plain Zone of Uttar Pradesh

Micronutrient Status	Percentage (%)	Macronutrient/Soil Property Status	Percentage (%)
Copper—Sufficient	94.47	Nitrogen—High	0.21
Copper—Deficient	5.53	Nitrogen—Medium	0.06
Boron—Sufficient	76.26	Nitrogen—Low	99.73
Boron—Deficient	23.74	Phosphorus—High	33.78
Sulfur—Sufficient	96.44	Phosphorus—Medium	62.29
Sulfur—Deficient	3.56	Phosphorus—Low	3.93
Iron—Sufficient	63.57	Potassium—High	0.19
Iron—Deficient	36.43	Potassium—Medium	60.88
Zinc—Sufficient	58.92	Potassium—Low	38.93
Zinc—Deficient	41.08	OC—High	0.09
Manganese—Sufficient	94.28	OC—Medium	21.48
Manganese—Deficient	5.72	OC—Low	78.43
		EC—Saline	0.09
		EC—Non-saline	99.91
		pH—Acidic	0.10
		pH—Neutral	99.72
		pH—Alkaline	0.18

Note: The values represent the percentages of soil samples classified into the respective categories. Micronutrients are classified as "Sufficient" or "Deficient," whereas nitrogen, phosphorus, potassium, and OC are classified as "High," "Medium," or "Low." EC and pH are classified according to the categories shown in the table. The reported percentages represent proportions of analyzed soil samples rather than direct nutrient concentrations. OC = organic carbon; EC = electrical conductivity. Data source: Ministry of Agriculture and Farmers Welfare, Government of India (n.d.).

The assessment is based on the targeted-yield equations for the National Capital Region of Delhi (AICRP on Soil Test Crop Response Correlations, n.d.), where F_{N}, $F_{\mathrm{P_2O_5}}$, and $F_{\mathrm{K_2O}}$ denote the recommended quantities (kg/ha) of nitrogen, phosphorus, and potassium, respectively; T represents the target yield (q/ha , where q denotes a quintal equivalent to 100 kg); S_{N}, S_{P}, and S_{K} denote the soil-available nitrogen, phosphorus, and potassium (kg/ha), respectively; and FYM represents the amount of farmyard manure applied (t/ha):

$$F_{\mathrm{N}} = 3.85T - 0.41S_{\mathrm{N}} - 1.64\mathrm{FYM}$$
$$F_{\mathrm{P_2O_5}} = 2.78T - 4.12S_{\mathrm{P}} - 1.72\mathrm{FYM}$$
$$F_{\mathrm{K_2O}} = 2.04T - 0.29S_{\mathrm{K}} - 0.88\mathrm{FYM}$$

(e) Seed sowing

Sustainable seed-sowing practices include allowing the soil to remain unplanted for at least 30 days before sowing, where appropriate, to facilitate soil recovery; sowing seeds at an appropriate depth to promote effective germination and access to moisture and nutrients; using cover crops to reduce soil erosion, suppress weeds, and increase organic matter; adopting conservation tillage to preserve soil quality and reduce erosion; and applying natural or biological pest-management strategies.

DBW-17 seed sowing: Soil moisture levels were determined to be acceptable; certified, pure seed was used; sowing was planned to improve field efficiency; and seed was line-sown at a rate of 100 kg/ha, with row spacing of 18–20 cm and a sowing depth of 5 cm.

(f) Irrigation

Irrigation is an important activity throughout the crop lifecycle and is closely related to SDG 6 (Clean Water and Sanitation), particularly in terms of reducing water pollution and improving water-use efficiency. Recommended strategies include the use of micro-irrigation technologies, such as sprinklers and drip irrigation, which can reduce water consumption and improve productivity. Targeting water application to the root zone can also substantially reduce water losses compared with flood irrigation. Rainwater harvesting can supplement irrigation, particularly in areas with irregular rainfall. Advanced irrigation systems incorporating soil-moisture sensors, meteorological data, and scheduling algorithms can optimize irrigation timing based on real-time conditions, thereby improving water-use efficiency and productivity. Solar-powered irrigation pumps can reduce dependence on fossil fuels, lower pollution, and decrease operating costs. In addition, cover crops and crop rotation can improve soil structure and enhance water retention.

Conceptual smart-irrigation framework: This study presents a conceptual framework for translating the aforementioned smart-irrigation techniques into practical technical applications. The framework is introduced as a design concept.

- Sensing layer. Soil-moisture sensors are positioned within the wheat root zone, together with an automated meteorological station that records precipitation, temperature, and humidity.
- Decision layer. A threshold- or rule-based scheduling algorithm evaluates real-time sensor data against crop-stage water requirements and provides an irrigation recommendation when soil moisture falls below the stage-specific threshold.
- Actuation layer. Drip or sprinkler valves are activated for the recommended irrigation event, replacing fixed-calendar irrigation with condition-based scheduling.
- Energy layer. Where practicable, pumps are powered by solar photovoltaic systems, reducing dependence on grid electricity or diesel.

DBW-17 irrigation schedule: Considering the alluvial soils of Western Uttar Pradesh, irrigation was applied at a depth of approximately 8 cm per event and scheduled at six key growth stages:

- First irrigation: 20–25 days after sowing (crown-root initiation stage).
- Second irrigation: 40–45 days after sowing (tillering stage).
- Third irrigation: 60–65 days after sowing (jointing stage).
- Fourth irrigation: 80–85 days after sowing (flowering stage).
- Fifth irrigation: 100–105 days after sowing (milk stage).
- Sixth irrigation: 120–125 days after sowing (dough stage).

Weather forecasts were evaluated before each irrigation event to improve water and field management.

(g) Harvesting

Sustainable harvesting practices aim to minimize environmental impacts while maintaining productive and efficient yields. Recommended practices include harvesting at the appropriate maturity stage, minimizing soil disturbance through reduced-tillage or no-till approaches, improving water-use efficiency during cultivation, recycling agricultural residues as compost or mulch to enhance soil health and reduce erosion, and adopting harvesting methods that minimize crop and soil damage.

Application to DBW-17 wheat: The DBW-17 variety has a harvest period of approximately 125–130 days after sowing. Daily and weekly weather forecasts were monitored from day 110 onward. A reaper was used for harvesting because of its suitability for medium-sized farms, faster operation, and cost efficiency.

(h) Storage

Proper grain-storage practices are essential for reducing post-harvest losses and maintaining wheat quality. Recommended practices include adequately drying harvested grain before storage, maintaining appropriate moisture levels, using clean and sanitized storage containers, protecting grain from pests and microbial contamination, and controlling temperature and humidity where possible. Appropriate packaging, handling, and storage conditions should also be adopted to minimize deterioration and preserve grain quality during the storage

period.

Application to DBW-17 wheat: The storage process was conducted in three stages.

Phase 1—Preparation: The harvested grain had a moisture content of approximately 15–20%. It was subsequently sun-dried for about one week and cleaned to remove soil, plant residues, and other foreign materials.

Phase 2—Storage: The cleaned grain was stored in sanitized metal bins and bags.

Phase 3—Post-storage pest control: Dried neem leaves were used as a natural pest deterrent, small sachets of slaked lime were placed in the storage containers to absorb excess moisture, and fumigation was applied when necessary.

The principles of sustainable agriculture emphasize meeting the needs of the present population without compromising the ability of future generations to meet their own needs, while maintaining economic viability, environmental sustainability, and social equity. The practices described above are intended to improve food security, enhance resilience to changing climatic conditions, and reduce the adverse environmental impacts associated with conventional farming.

3. Results and Discussion

The growing population of developing countries and the increasing risk of food insecurity have stimulated continued research into sustainable agricultural practices. This study synthesizes recognized sustainable and organic farming practices across the major stages of the crop lifecycle and demonstrates their application through an illustrative wheat case study in Western Uttar Pradesh.

The principal outcome of this study is a qualitative mapping of the proposed sustainable-practice framework to relevant SDGs. Table 3 directly associated with specific practices within the proposed framework, whereas Table 4 identifies SDGs that may be indirectly influenced through broader effects on agricultural productivity, income, resource use, and environmental quality. These indirect relationships are conceptual and were not empirically evaluated in the present study.

Table 3. Sustainable Development Goals (SDGs) directly associated with the proposed framework

SDGs	Supporting Practices	Rationale
SDG 2—Zero Hunger	Crop selection, seed selection, nutrient management, and water management	Improves the efficiency and stability of wheat production, thereby contributing to food security.
SDG 6—Clean Water and Sanitation	Micro-irrigation, smart irrigation scheduling, and organic fertilization	Improves water-use efficiency and may reduce agrochemical leaching into aquatic environments.
SDG 7—Affordable and Clean Energy	Solar photovoltaic irrigation pumps	Reduces dependence on diesel- or grid-powered irrigation systems through the use of renewable energy.
SDG 12—Responsible Consumption and Production	Use of biofertilizers and compost, residue reuse, and reduced synthetic inputs	Promotes efficient resource use and reduces agricultural waste.
SDG 13—Climate Action	Reduced use of synthetic fertilizers, conservation tillage, and solar-powered irrigation	May reduce greenhouse gas emissions and energy consumption associated with agricultural production.
SDG 15—Life on Land	Crop rotation, diversified seed selection, and organic amendments	Supports soil health and agricultural biodiversity.

Table 4. Sustainable Development Goals (SDGs) indirectly associated with the proposed framework

SDGs	Rationale
SDG 1—No Poverty	Improved and more stable agricultural income may contribute to poverty reduction if the proposed practices increase or stabilize farm productivity.
SDG 8—Decent Work and Economic Growth	The proposed practices may contribute indirectly to agricultural employment, farm income, and regional economic activity.
SDG 9—Industry, Innovation and Infrastructure	The proposed smart-irrigation framework may support agricultural technological innovation if implemented in practice.
SDG 11—Sustainable Cities and Communities	Potential indirect contributions may arise through improved rural–urban food-system linkages and resource management.
SDG 14—Life Below Water	Reduced agrochemical runoff may decrease downstream nutrient loading and associated impacts on aquatic ecosystems.
SDG 17—Partnerships for the Goals	Implementation may depend on cooperation among farmers, extension services, government agencies, researchers, and other stakeholders.

The mapping indicates that irrigation and fertilization practices are primarily associated with water-, energy-,

and climate-related goals (SDGs 6, 7, and 13), whereas crop and seed selection, sowing, and storage are more closely associated with food security, responsible resource use, and terrestrial ecosystem objectives (SDGs 2, 12, and 15). This distinction separates SDGs that are directly linked to specific practices within the proposed framework from those that may be influenced indirectly through broader socioeconomic or environmental pathways. These relationships remain conceptual and were not empirically tested in the present study.

4. Conclusion

Agriculture plays an important role in sustainable development because agricultural practices directly affect food security, water resources, soil quality, biodiversity, energy use, and greenhouse gas emissions. Sustainable practices such as soil conservation, efficient water management, reduced reliance on synthetic inputs, biodiversity protection, and the adoption of renewable-energy technologies may help reduce the environmental impacts of agricultural production.

In this study, a conceptual framework integrating sustainable and organic farming practices across the crop lifecycle was developed and illustrated through a wheat case study in Western Uttar Pradesh. The framework was further mapped to SDGs that are directly or indirectly associated with these practices. Because the present study is conceptual and relies primarily on secondary information rather than field-based experimental validation, future research should evaluate the proposed framework using quantitative field data to determine its actual agronomic, environmental, and socioeconomic effects.

Author Contributions

Conceptualization, S.T. and Y.K.G.; methodology, S.T. and Y.K.G.; software, S.T. and Y.K.G.; validation, S.T. and Y.K.G.; formal analysis, S.T. and Y.K.G.; investigation, S.T. and Y.K.G.; resources, S.T. and Y.K.G.; data curation, S.T. and Y.K.G.; writing—original draft preparation, S.T. and Y.K.G.; writing—review and editing, S.T. and Y.K.G.; visualization, S.T. and Y.K.G.; supervision, Y.K.G.; project administration, Y.K.G. All authors have read and agreed to the published version of the manuscript.

Data Availability

No primary data were collected. The secondary data sources are cited in the manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

References

AICRP on Soil Test Crop Response Correlations. (n.d.). *Four Decades of STCR Research: Crop-Wise Recommendations*. Indian Institute of Soil Science, Bhopal, India. https://iiss.res.in/downloads/division/stcr%20Crop%20wise%20Recommendations.pdf

Asiedu-Ayeh, L. O., Zheng, X., Agbodah, K., Dogbe, B. S., & Darko, A. P. (2022). Promoting the adoption of agricultural green production technologies for sustainable farming: A multi-attribute decision analysis. *Sustainability*, *14*(16), 9977. https://doi.org/10.3390/su14169977.

Chimi, P. M., Mala, W. A., Tchonang Djoumbi, B. L., Fobane, J. L., Fankam Kenmoe, F. U., Ngamsou Abdel, K., Yonga, G., Ngon Dikoume, A. M., & Bell, J. M. (2025). Indigenous agroecological knowledge and practices for climate change adaptation in the forest-savanna transition zone at Batchenga in the Centre Region of Cameroon. *Discov. Environ.*, *3*, 16. https://doi.org/10.1007/s44274-025-00192-z.

Chiriacò, M. V., Grossi, G., Castaldi, S., & Valentini, R. (2017). The contribution to climate change of the organic versus conventional wheat farming: A case study on the carbon footprint of wholemeal bread production in Italy. *J. Clean. Prod.*, *153*, 309–319. https://doi.org/10.1016/j.jclepro.2017.03.111.

Hiywotu, A. M. (2025). Advancing sustainable agriculture for Goal 2: Zero Hunger – A comprehensive overview of practices, policies, and technologies. *Agroecol. Sustain. Food Syst.*, *49*(7), 1027–1055. https://doi.org/10.1080/21683565.2025.2451344.

ICAR–Indian Institute of Wheat and Barley Research. (n.d.). *Varieties of wheat*. https://iiwbr.icar.gov.in/varieties-of-wheat/

Ikram, M., Ferasso, M., Sroufe, R., & Zhang, Q. (2021). Assessing green technology indicators for cleaner production and sustainable investments in a developing country context. *J. Clean. Prod.*, *322*, 129090. https://doi.org/10.1016/j.jclepro.2021.129090.

Kiełbasa, B., Pietrzak, S., Ulén, B., Drangert, J. O., & Tonderski, K. (2018). Sustainable agriculture: The study on

farmers' perception and practices regarding nutrient management and limiting losses. *J. Water Land Dev.*, *36*(1), 67–75. https://doi.org/10.2478/jwld-2018-0007.

Kumar, S. & Manshi. (2023). Sustainable agriculture development in India: Emerging issues, challenges and opportunities. *Asian J. Adv. Agric. Res.*, *23*(3), 64–74. https://doi.org/10.9734/ajaar/2023/v23i3469.

Lee, H. J. & Yun, Z. S. (2015). Consumers' perceptions of organic food attributes and cognitive and affective attitudes as determinants of their purchase intentions toward organic food. *Food Qual. Prefer.*, *39*, 259–267. https://doi.org/10.1016/j.foodqual.2014.06.002.

Ministry of Agriculture and Farmers Welfare, Government of India. (n.d.). *Soil Health Card Portal: Nutrient Dashboard*. https://soilhealthcard.krishitantra.com/home

Page, K. L., Dang, Y. P., & Dalal, R. C. (2020). The ability of conservation agriculture to conserve soil organic carbon and the subsequent impact on soil physical, chemical, and biological properties and yield. *Front. Sustain. Food Syst.*, *4*, 31. https://doi.org/10.3389/fsufs.2020.00031.

Rizzo, G., Migliore, G., Schifani, G., & Vecchio, R. (2024). Key factors influencing farmers' adoption of sustainable innovations: A systematic literature review and research agenda. *Org. Agr.*, *14*(1), 57–84. https://doi.org/10.1007/s13165-023-00440-7.

Scherer, S. F., Wecker, K., Schneider, R., Hanano, A., Fitzgerald, G., Vaughan, A., Ní Chéilleachair, R., Mann, H., Weller, D., Radtke, K., et al. (2025). *2025 Global Hunger Index: 20 Years of Tracking Progress: Time to Recommit to Zero Hunger*. Concern Worldwide, Welthungerhilfe, & Institute for International Law of Peace and Armed Conflict (IFHV). https://www.globalhungerindex.org/pdf/en/2025.pdf

Schnack, A., Bartsch, F., Osburg, V., & Errmann, A. (2024). Sustainable agricultural technologies of the future: Determination of adoption readiness for different consumer groups. *Technol. Forecast. Soc. Change*, *208*, 123697. https://doi.org/10.1016/j.techfore.2024.123697.

Suji, D. B. & Kumar, A. M. S. (2020). A study on adoption of eco-friendly technologies and constraints. *Plant Arch.*, *20*(1), 42–46.

Tyagi, S. & Gupta, Y. K. (2026). An intelligent smart farming decision support system for sustainable crop lifecycle management and SDG 2 achievement. *Int. J. Drug Deliv. Technol., 16*(56s), 1489–1505. https://doi.org/10.25258/ijddt.16.56s.161.

United Nations. (2019). *Food*. https://www.un.org/en/global-issues/food

United Nations Economic and Social Commission for Western Asia. (2018). *The 5Ps of the Sustainable Development Goals*. https://www.unescwa.org/sites/default/files/inline-files/the_5ps_of_the_sustainable_development_goals.pdf

Vats, S. (2025). Sustainable agriculture practice: A review of emerging policies and practices in India. *Lex Localis J. Local Self-Gov.*, *23*(S6), 919–939. https://doi.org/10.52152/801883.